

Dhyey Patel

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EDUCATION

Eastern Michigan University (M.S. in Computer Science | GPA: 4.0/4.0)

Expected Dec 2026

University of Cincinnati (B.S. in Computer Science with minor in Math | GPA: 3.744/4.0)

Aug 2020 – May 2025

EXPERIENCE

Eastern Michigan University - Ypsilanti, MI

Web Developer Graduate Assistant (Part-time)

Sep 2025 – Present

- Engineered and launched **3 WCAG-friendly transfer-planning tools** by transforming articulation-guide data into structured JSON and building accessible 2-step program discovery workflows, replacing static link directories for transfer students across partner colleges.
- Standardized **Request More Information** workflows across **500+** academic program pages and 6 colleges by centralizing form logic, recipient mappings, and program metadata, driving **~100% year-over-year growth** in inquiry submissions.
- Reworked catalog-link infrastructure across **400+ pages**, replacing a recurring manual update process with a more durable implementation that reduced annual maintenance from **~2 weeks of team effort to ~1 day for 2 people**.

Givaudan - Cincinnati, OH

Frontend Developer Intern (Co-op)

Apr 2023 – Aug 2023

- Rebuilt **3 ATOM pages in React + MUI**, cutting page build time from **~2 weeks to ~3 days**.
- Converted an **R/MATLAB** algorithm to **JavaScript**, reducing runtime from **~5s to <0.5s** with output parity preserved.
- Fixed production issues in Insight's data-retrieval table and shipped the final updates within **1 week**, improving a workflow used by internal teams globally.

Web Developer Intern (Co-op)

Aug 2022 – Dec 2022

- Built internal experiment-creation and response-capture tools for Givaudan's Insight platform using **HTML**, **CSS**, and **JavaScript**, including a reusable numpad, dynamic image/GIF forms, and **Base64-based media storage** for reliable retrieval.
- Engineered a **Flask-based admin dashboard** with searchable tables, pagination, statistics, and charts to review submitted test data asynchronously, reducing retrieval time to **<1s**.
- Developed interactive **D3.js** visualizations—including bidirectional bar, radar/spider, and donut charts—that replaced slower legacy visuals in ATOM 2.0 and cut result-view load times from **up to 1 minute to <3 seconds**.

PROJECTS

SegLungAI ([GitHub](#)) | PyTorch, scikit-learn, NumPy, U-Net, ResNet

Aug 2024 – Apr 2025

- Built an end-to-end **neonatal lung MRI segmentation pipeline** covering preprocessing, training, evaluation, and inference workflows.
- Evaluated performance using IoU/Dice, achieving **~0.90 IoU**, and packaged reproducible training scripts for repeatable experimentation.

BirdCLEF 2026 ([Kaggle](#)) | PyTorch, timm, librosa, EfficientNet-B2

Mar 2026 – Present

- Built a multilabel bird-audio classification pipeline using **mel spectrograms**, EfficientNet-B2, **weighted sampling**, and SpecAugment.
- Trained and validated on focal clips and soundscapes, achieving the best **validation macro-AUC of 0.8725** on a **234-class** setup.

Drone Flight Data Analytics ([GitHub](#)) | Kafka, Spark, AWS EC2/S3

Oct 2025 – Nov 2025

- Built a telemetry analytics pipeline on AWS using Kafka for ingestion, Spark for processing, and S3 for persisted outputs.
- Processed flight streams into structured datasets and automated repeatable runs for telemetry analysis and review.

Job Fair Match ([Live](#)) | Next.js, TypeScript, Prisma, PostgreSQL, NeonDB, Vercel

Sep 2025 – Dec 2025

- Built a swipe-based job matching platform with responsive browse → swipe → match flows and production-ready routing/state handling.
- Designed a normalized PostgreSQL schema and backend APIs for profiles, swipes, and matches, enforcing correctness with constraints and fast lookups.

Automata Workbench ([Live](#)) | JavaScript, Algorithms, Interactive UI

Oct 2025 – Nov 2025

- Implemented core automata algorithms (NFA to DFA conversion, DFA minimization) and validated correctness with test cases.
- Built an interactive UI to input automata and visualize states/transitions to support debugging and learning.

SKILLS

Languages: Python, TypeScript/JavaScript, SQL, C++, R, VBA

ML: PyTorch, scikit-learn, NumPy, pandas, librosa, timm

Web: React, Next.js, Node.js, Express, Flask/Django, REST APIs, MUI, D3.js

Data: PostgreSQL, MongoDB, Prisma, Kafka, Spark, AWS (EC2, S3), Azure, Docker, Linux, Git, GitHub Actions

AWARDS

Graduate Assistantship (Full Tuition Waiver)

Sep 2025

UC Global Scholarship, International Outreach Scholarship, Bearcats Everywhere Scholarship

Nov 2020

Dean's List (Fall 2020 - Spring 2025)

Feb 2021